

Tawyabul Islam Tamim

B.Sc. (Honours), 4th-Year Student
Department of Computer Science and Engineering
University of Dhaka
Dhaka, Bangladesh

 tamim04.github.io

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Education

Bachelor of Science

Sept 2022 – Expected Feb 2027

Department of Computer Science and Engineering, University of Dhaka, Dhaka, Bangladesh

CGPA: 3.45/4.00 (through 7th semester)

Undergraduate Thesis (Ongoing): “ARGUS: An Adaptive, Resource-Aware, Grounded Understanding System for Real-Time Explainable Video Anomaly Detection” — Developing a resource-aware framework for real-time video anomaly detection with selective vision-language reasoning and evidence-grounded explanations.

Supervisor: [Md. Ahasanul Alam](#)

Research Interests

Deep Learning & Foundation Models, Computer Vision & Multimodal AI, AI Agents & Autonomous Systems, AI for Climate & Earth System Science, Robotics & Physical AI.

Research Projects

BD-HeatPred: A Multi-Source Daily Predictor Dataset for Heatwave Research in Bangladesh, February–June, 1991–2025

Ongoing

- Developing a region-specific dataset for Bangladesh by integrating 35 atmospheric, land-surface, and global teleconnection predictors across five regions: Dhaka, Khulna, Rajshahi, Rangpur, and Rangamati–Bandarban–Khagrachari (RBK).

Supervisor: [Shuchi Chaki](#)

BD-HeatPred-Based Heatwave Onset Prediction and Regional Driver Analysis in Bangladesh

Ongoing

- Predicting heatwave onset in Bangladesh using the BD-HeatPred dataset and analyzing regional drivers to identify the most influential predictors across five regions.

Supervisor: [Shuchi Chaki](#)

Projects

Deep Learning-Based Illegal Road Crossing Detection in Dhaka Urban Traffic Surveillance

 [Code](#)

- Created a Dhaka-specific pedestrian dataset from six locations under diverse conditions.
- Introduced a spatially aware 3-class scheme: person_middle, person_side, and person_sidewalk.
- Trained and compared YOLOv8m/l and YOLO11m/l on single and multi-class detection tasks.
- Conducted 640/960/1280 px resolution ablation, selecting 960×960 as the practical operating point.

Tools: Python, PyTorch, Ultralytics YOLO, Roboflow

ATM Simulator System

 [Code](#)

- Developed a Java Swing-based ATM simulator with secure card/PIN authentication and complete banking workflows including account registration, deposits, withdrawals, fast cash, PIN changes, and balance enquiry.

Tools: Java, Java Swing, JDBC, MySQL, Apache Ant

FamNest — Family Finance & Organizing App

[Code](#)

- Contributed to FamNest, a family management application for expense tracking, document storage, event management, and digital memory preservation.
- Developed the expense tracking module, enabling users to record, organize, and manage family expenses.

Tools: Flutter, Dart, FastAPI, MongoDB Atlas, Cloudinary

Design and Implementation of an Amplitude Modulator and Demodulator

[Code](#)

- Designed and implemented transistor/diode-based AM modulator and demodulator circuits, fabricating both on custom PCBs.
- Tested signal transmission and recovery using an oscilloscope, analyzing the modulated and demodulated waveforms and overall modulation behavior.

Tools: PCB Design and Fabrication, Analog Circuits, Signal Generator, Oscilloscope, Multimeter, Soldering

⚙️ Skills and Tools

Programming	Python, C, C++, Java
Machine Learning & Deep Learning	PyTorch, Keras, NumPy, Pandas, Matplotlib, Seaborn, Hugging Face
Computer Vision	OpenCV, Roboflow, Ultralytics YOLO
Climate & Geospatial Analysis	ERA5/ERA5-Land, NetCDF, Xarray, GeoPandas, GIS, Spatial Analysis, Grid-Based Analysis, Remote Sensing (NDVI, NDWI), Google Earth Engine (GEE), Climate Anomaly & Teleconnection Analysis (SST, ENSO, IOD, MJO, OLR)
Web/Software Development	React.js, Node.js, Express.js, Java Swing
Databases	MySQL, PostgreSQL, MongoDB, Oracle Cloud Database
Embedded Systems	ESP32, STM32F446RE, Keil uVision, Arduino
Development & Research Tools	Git, GitHub, Linux, VS Code, PyCharm, LaTeX, Google Colab, Jupyter Notebook
Networking & Design Tools	Cisco Packet Tracer, Canva

🏆 Competitive Programming

200+ problems solved on [Codeforces](#), [LeetCode](#), [CSSES](#), [CodeChef](#) and [HackerRank](#), mainly covering implementation, mathematics and number theory, greedy algorithms, strings, sorting and searching, graph traversal (DFS/BFS), and core data structures.

★ Scholarship

Awarded the General Scholarship for outstanding academic performance in the Secondary School Certificate (10th Grade) Examination, Dhaka Board (2019).

Standardized Tests

I plan to take the **International English Language Testing System (IELTS)** and the **Graduate Record Examination (GRE)** soon.

References

Md. Ahasanul Alam

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University of Dhaka, Dhaka, Bangladesh
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Shuchi Chaki

Graduate Teaching Assistant, North Carolina State University
Lecturer (on study leave), Department of Applied Mathematics
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Kawser Ahammed

PhD Student (fifth year), Department of Electrical and Computer Engineering
Old Dominion University, Virginia, USA
Assistant Professor (on study leave)
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